Medical Imaging Notes

ELC5396
Spring 2025
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Contents

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1 Imaging Modalities & Breakpoints (Tuesday 1/21/2025)

- examples of medical imaging: MRI, Ultrasound(US), XRay, CT, PET, Mass Spec, SPECT, DOT, anger cameras
- a modality: danger vs safe. what's a breakpoint?
  - use 13.6 eV: energy in UV light
  - This is the energy needed for an electron in a single H atom to break free; we call this *ionizing energy*. Since the human body is mostly water (and thus mostly hydrogen), this is roughly the minimum energy to ionize the body's atoms. Once ionized, the freed atoms (e.g. oxygen) can bond with other molecules; if too many bonds break, the cell dies as the ionized oxygen damages DNA. So 13.6 eV is the breakpoint for whether radiation can ionize cells, and thus whether it can damage DNA.
- Another break point: how to take an image: three times
  - **transmission**: source outside, receiver on other end
  - **reflective**: source and receiver outside, same location
  - **emission**: source inside object receiver outside

2 Convolution, Systems & Linear Properties (Thursday 1/23/2025)

- Signals and systems, what's the thought?
  - *convolutions*: allow us to determine system characteristics.
  - simply "multiplication"
- **Convolution**: $*$ operator
  - $h(n) = (f * g)(n) = \sum_m f(m)g(n-m)$

    - the $-m$ is the reflection, n is the shift in time
    - integration for continuous function
    - important takeaway: we're worrying about many shifts (which correspond with the index)
- If our example is $[1,2,1] * [2,1] = [2,5,4,1]$
    - Another way of interpretation?
    - polynomial multiplication; $(1+2x+x^2)(2+x) = 2+5x+4x^2+x^3$
    - makes for easy way to perform polynomial multiplication!
- $O(n^2)$ operation
- for medical images, each pixel considered its own dimension, so operation complexity is important
- instead of performing convolution which is $O(n^2)$, maybe we can take FT since we like working in multiplication
- $\mathcal{F}\{f(t) * g(t)\} = F(u)G(u)$ via the fourier transform
    - from convolution to point-by-point multiplication or *array multiplication*
    - So we go from $O(n^2)$ to $O(n)$ for the operation complexity
    - but we have to account for FT calculation; this is $O(n^2)$
    - instead of basic FT, can use FFT, accelerates to $O(n \log n)$ by taking advantage of FT and splitting into binary tree
    - random fact, $\log(n)$ is the slowest growing function that still grows to infinity
- introduce emphasis on speed: we want fast performance
- Figure 1 for system in series
    - $g_0 = h_1 * f$ fed into H_2 so $g = (h_2 * h_1) * f$
    - can turn into one big box: $G = H_1 H_2 F$
    - this setup is in *series*

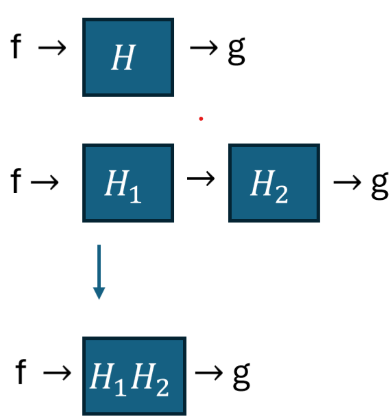

Figure 1: Systems in Series

- Figure 2 parallel system
    - Other setup in *parallel*

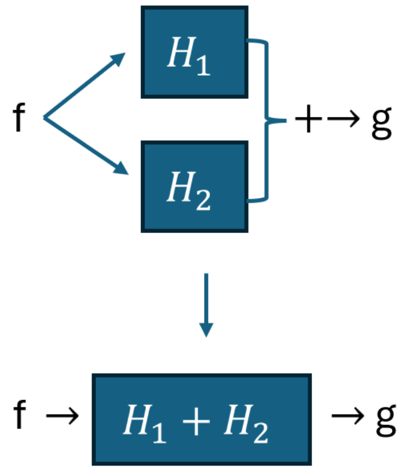

Figure 2: Systems in Parallel

    * $g = h_1 * f + h_2 * f = (h_1 + h_2) * f$

- Some systems we like: *LTI (time invariant)* or *LSI (space invariant)*; the difference is in the variable we deal with. We typically work with shift invariant systems in this course, and rarely deal with time only. Shifts can be in space and time.
- some other helpful functions and properties
    - Say we want to calculate the FT of an image. In general, we say it's $O(n^2)$
    - if our image is n by n, then the FT is $O(n^4)$
    - instead of FT in both directions at once, we can simplify to FT in x direction then in y direction due to separability of the FT operation
- so far, we have three important properties: linear, shift invariant, and separable.
    - if we ask whether a system is *linear*, we check for the *additive* and *scaling* (homogeneity) properties
    - checking for *shift invariant*, we check for $G(f(x + \Delta x)) = g(x + \Delta x)$